CHEMISTRY

1) Which of the following order is correct -

(1) Basic character : $Li_2O > Na_2O > K_2O > Cs_2O$
(2) Acidic character : $Al_2O_3 < SiO_2 < P_2O_5 < Cl_2O_7$
(3) Acidic strength : HF > HCl > HBr > HI
(4) Acidic character : $N_2O > NO_2 > N_2O_5 > N_2O_3$

2) Which of the following order of atomic radius is correct :-

(1) Sc < Y ≈ La
(2) K < Rb ≈ Cs
(3) Ge < Sn > Pb
(4) Ni < Pd ≈ Pt

3) Which element shows both positive and negative oxidation states :

(1) Cs
(2) F
(3) I
(4) Xe

4) Which of the following species is paramagnetic and have bond order equal to F_2 ?

(1) H_2
(2) O_2
(3) B_2
(4) Li_2

5) Which of the following order is incorrect ?

(1) Ionic character : $MCl < MCl_2 < MCl_3$
(2) Polarisibility : $F^- < Cl^- < Br^- < I^-$
(3) Polarising power : $Na^+ < Mg^{+2} < Al^{+3}$
(4) Covalent character : LiF < LiCl < LiBr < LiI

6) Which of the following have highest bond length?

(1) O–H
(2) C–H

(3) N–O
(4) C–O

7) Which of the following molecule has highest bond energy ?

(1) F — F
(2) N — N
(3) C — C
(4) O — O

8) The covalency of nitrogen in HNO_3 is :-

(1) 0
(2) 3
(3) 4
(4) 5

9) Which one of the following arrangements of molecules is correct on the basis of their dipole moments ?

(1) $BF_3 > NF_3 > NH_3$
(2) $NF_3 > BF_3 > NH_3$
(3) $NH_3 > BF_3 > NF_3$
(4) $NH_3 > NF_3 > BF_3$

10) Complexes $[Pt(NH_3)_4Cl_2]Br_2$ and $[Pt(NH_3)_4Br_2]Cl_2$ are examples of which type of isomerism ?

(1) Ionisation Isomerism
(2) Linkage isomerism
(3) Hydrate isomerism
(4) Co-ordination isomerism

11) Which of the following is square planar and paramagnetic complex?

(1) $[NiCl_4]^{-2}$
(2) $[Pt(CN)_4]^{-2}$
(3) $[Zn(OH)_4]^{-2}$
(4) $[Cu(NH_3)_4]^{2+}$

12)

The IUPAC name of the complex $[Ag(H_2O)_2][Ag(CN)_2]$ is :

(1) diaquasilver(II) dicyanidoargentate(II)
(2) dicyanidosilver(I) diaquaargentate(I)
(3) diaquasilver(I) dicyanidoargentate(I)

(4) dicyanidosilver(II) diaquaargentate(II)

13) Which has maximum boiling point :-

(1) Ne
(2) Ar
(3) Xe
(4) He

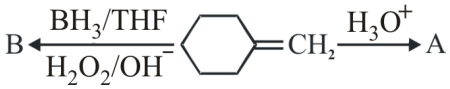
14) Which of following halide is inert towards hydrolysis at room temperature :-

(1) $SiCl_4$
(2) PCl_3
(3) NF_3
(4) NCl_3

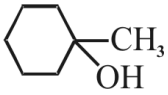
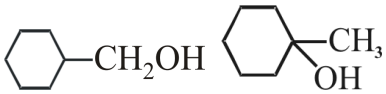
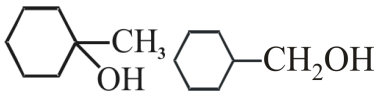
15) $CrO_4^{2-} \underset{pH=Y}{\overset{pH=X}{\rightleftharpoons}} Cr_2O_7^{-2}$

Values of (X) and (Y) are respectively

(1) 4 and 5
(2) 4 and 8
(3) 8 and 4
(4) 8 and 9

16) 

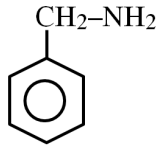
A and B are :-

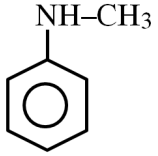
(1) Both 
(2) Both 
(3) 
(4) 

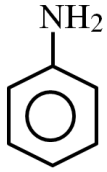
17) Which of the following reacts with Hinsberg reagent to give product which is soluble in alkali?

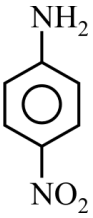
(1) $CH_3–CH_2–NH_2$
(2) $(CH_3CH_2)_2NH$
(3) $(CH_3CH_2)_3N$
(4) $\overset{\oplus}{N}(CH_3CH_2)_4\,Cl^{\ominus}$

18) Which of the following is the strongest base :-

(1) 

(2) 

(3) 

(4) 

19) Ethyl phenyl ether on boiling with cold con. HBr gives

(1) phenol and ethane

(2) bromo benzene and ethane

(3) phenol and ethyl bromide

(4) bromobenzene and ethanol

20) The compound A, B and C in the reaction sequence

$CH_3–CH_2OH \xrightarrow{PBr_3} A \xrightarrow{Alc.KOH} B \xrightarrow{Br_2} C$

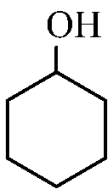
are given by the set :-

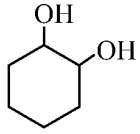
(1) C_2H_5Br, CH_3CH_2OH, CH_3CHBr_2

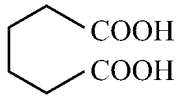
(2) C_2H_5Br, $CH{\equiv}CH$, $CH_2{=}CHBr$

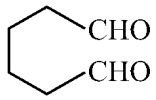
(3) C_2H_5Br, $CH_2{=}CH_2$, $Br–CH_2–CH_2–Br$

(4) C_2H_5Br, CH_3CH_2OH, $Br–CH_2–CH_2–Br$

21)  $\xrightarrow[\Delta]{H_2SO_4}$ (A) $\xrightarrow[\Delta]{KMnO_4/H^+}$ (B) Product B is :-

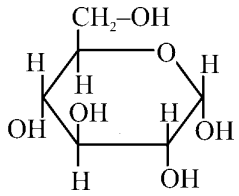
(1) 

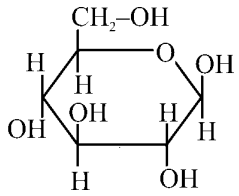
(2) 

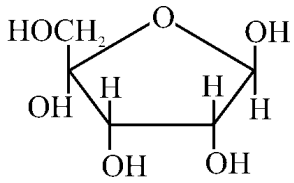
(3) 

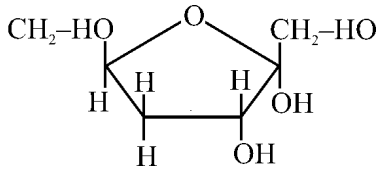
(4) None

22) Which of the following is structure of β-D-Glucopyranose ?

(1) 

(2) 

(3) 

(4) 

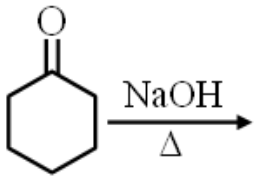
23) Reaction of carbonyl compound with one of the following reagents involves nucleophilic addition followed by elimination of water. The reagent is :

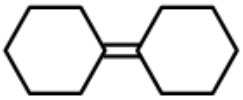
(1) hydrocyanic acid (HCN)

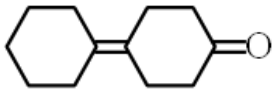
(2) sodium hydrogen sulphite ($NaHSO_3$)

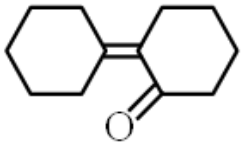
(3) a Grignard reagent

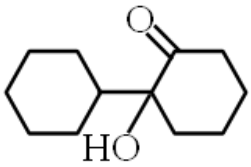
(4) hydrazine in presence of feebly acidic solution

24)  (A) . (A) is?

(1) 

(2) 

(3) 

(4) 

25) ***Assertion :-*** O-nitrophenol is more acidic than m-nitrophenol.
Reason :- Intramolecular H-bonding is present in O-nitrophenol.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

26) **Assertion :** Staggered conformation of ethane is most stable while eclipsed conformation is least stable.
Reason : Staggered form has the least torsional strain and the eclipsed form has the maximum torsional strain

(1) Assertion is correct, reason is correct, reason is a correct explanation for assertion

(2) Assertion is correct, reason is correct, reasion is not a correct explanation for assertion

(3) Assertion is correct, reason is incorrect.

(4) Assertion is incorrect, reason is correct.

27) **Statement-I :** Cross aldol condensation of ethanal and propanal gives mixture of four structural products (excluding stereoisomers).
Statement-II : Ethanal and propanal, both have $\alpha - H$.

(1) Statement I is correct and Statement II is incorrect

(2) Statement I is incorrect and Statement II is correct

(3) Both Statement I and Statement II are correct

(4) Both Statement I and Statement II are incorrect

28) Which purification technique is used for if the difference in boiling points of two liquids is very low.

(1) Simple distillation

(2) Steam distillation

(3) Fractional distillation

(4) Reduced pressure distillation

29) Consider the following statements about chromatography.
(a) Silica gel can be used as an adsorbent in column chromatography.
(b) Thin layer chromatography is a type of adsorption chromatography.

(c) In paper chromatography, the chromatography paper contains water molecules trapped in it.
the correct statement are -

(1) a & b only
(2) b & c only
(3) a & c only
(4) a, b & c

30) Which of the following ether cannot be prepared by williamson ether synthesis :-

(1) Methyl phenyl ether
(2) Methyl t-butyl ether
(3) Diethyl ether
(4) Diphenyl ether

31) If the de-broglie wave length of an electron revolving in 2nd orbit of H-atom is x. Then radius of the orbit is :-

(1) $\frac{x}{\pi}$
(2) $\frac{2x}{\pi}$
(3) $\frac{x}{2\pi}$
(4) $\frac{4x}{\pi}$

32) If the electron can be located in the orbit within 0.1Å. What is the uncertainty in its velocity :-

(1) $5.76 \times 10^{7}\ ms^{-1}$
(2) $0.576 \times 10^{7}\ ms^{-1}$
(3) $0.576 \times 10^{-4}\ ms^{-1}$
(4) $5.76 \times 10^{-4}\ ms^{-1}$

33) When the pressure is applied over system,
ice $\rightleftharpoons$ water, at equilibrium what will happen :-

(1) There will be no effect over equilibrium.
(2) More ice will form.
(3) More water will form.
(4) Water will decompose in H_2 and O_2.

34) N_2O_4 is dissociated to 33% and 50% by mole at total pressure P_1 and P_2 atm respectively. The ratio of P_1/P_2 is :

(1) 7/4
(2) 7/3

(3) 8/3

(4) 8/5

35) What is the concentration of Ba^{2+} when BaF_2 (K_{sp} = 1.0 × 10^{-6}) begins to precipitate from a solution that is 0.10 M F^- ?

(1) 9.0 × 10^{-7} M

(2) 3.3 × 10^{-5} M

(3) 10^{-4} M

(4) 3.0 × 10^{-7} M

36) 100 ml, 0.1 M CH_3COOH solution is mixed with 50 ml, 0.1 M NaOH solution, then pH of resulting solution will be :-
(pk_a = 4.7)

(1) 4.7

(2) 9.3

(3) 5.3

(4) 6.8

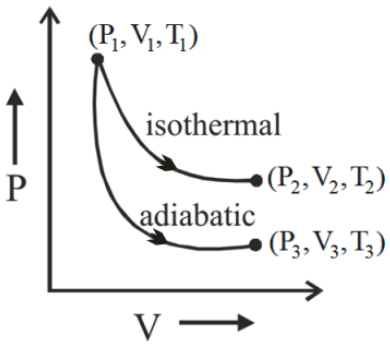
37) 2 mole of an ideal gas at 27°C expands reversibly from 2 litre to 20 litre. The entropy change is :
(R = 2 cal mol^{-1} K^{-1})

(1) 92.1

(2) zero

(3) 4

(4) 9.2



38) On the basis of curve, true statements are-

(1) $T_1 = T_2$

(2) $T_3 > T_1$

(3) $|W_{isothermal}| > |W_{adiabatic}|$

(4) $\Delta U_{isothermal} > \Delta U_{adiabatic}$

(1) 1, 3, 4

(2) 1, 2, 3

(3) 1, 2, 4

(4) 2, 3, 4

39) Which of the following is incorrect ?

(1) Colloidal sol of starch is an example of lyophilic sol.

(2) Ferric hydroxide sol is prepared by direct mixing it with water.

(3) Ferric hydroxide sol is lyophobic sol.

(4) Colloidal sol of egg albumin is prepared by mixing egg white in to H_2O.

40)

Standard electrode potentials of

$Fe^{2+} + 2e^- \rightarrow Fe$ and $Fe^{3+} + 3e^- \rightarrow Fe$ are –0.440 V and – 0.036 V respectively. The standard electrode potential E° for $Fe^{3+} + e^- \rightarrow Fe^{2+}$ is :-

(1) –0.476V

(2) –0.404V

(3) +0.404V

(4) +0.772V

41) One Faraday electricity is passed through the solutions of $AgNO_3$ and $CuSO_4$ connected in series. The mass of Ag and Cu deposited would be in the ratio :-
(Ag = 108, Cu = 63.5)

(1) 54 : 31.75

(2) 54 : 63.5

(3) 108 : 31.75

(4) 108 : 63.5

42)

For a first order homogeneous gaseous reaction $A \rightarrow 3B$, if pressure after time t was P_T and initial pressure was P^0 then select correct relation :-

(1) $k = \frac{1}{t} \ln \left[\frac{P^0}{3(P^0 - P_T)} \right]$

(2) $k = \frac{1}{t} \ln \left[\frac{2P^0}{3(P^0 - P_T)} \right]$

(3) $k = \frac{1}{t} \ln \left(\frac{3P^0}{2P^0 - P_T} \right)$

(4) $k = \frac{1}{t} \ln \left(\frac{2P^0}{3P^0 - P_T} \right)$

43) 99% of a first order reaction was completed in 32 minutes then 99.9% of the reaction will complete in :-

(1) 50 min

(2) 46 min

(3) 48 min

(4) 49 min

44) Which of the following forms the maximum boiling azeotrope ?

(1) n-hexane + n-heptane
(2) HNO_3 + H_2O solution
(3) C_2H_5OH + H_2O solution
(4) C_6H_6 + $C_6H_5CH_3$ solution

45) Mixture of volatile components A and B has total vapour pressure (in torr) p = 254 –119 x_A where x_A is mole fraction of A in liquid mixture. Hence p_A^o and p_B^o are (in torr) :-

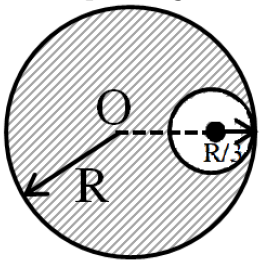
(1) 254, 119
(2) 119, 254
(3) 135, 254
(4) 119, 373

PHYSICS

1) Consider a uniform wire of mass 2M and length L. It is bent into a semi-circle. It's moment of inertia about a line perpendicular to the plane of the wire passing through the centre of circle, is :

(1) $\frac{ML^2}{\pi^2}$
(2) $\frac{ML^2}{2\pi^2}$
(3) $\frac{ML^2}{4\pi^2}$
(4) $\frac{2ML^2}{\pi^2}$

2) A disc has mass 9m. A hole of radius $\frac{R}{3}$ is cut from it as shown in the figure. The moment of inertia of remaining part about an axis passing through the centre 'O' of the disc and perpendicular to the plane of the disc will be :



(1) 8 mR^2
(2) 4 mR^2
(3) $\frac{40}{9}$ mR^2

(4) $\frac{37}{9}$ mR^2

3) A moving coil galvanometer has 100 equal division on it's scale. Its current sensitivity is 5div/mA and voltage sensitivity of 1 division per millivolt. In order that each division reads 1V, the resistance in ohm needed to be connected in series with its coil is :

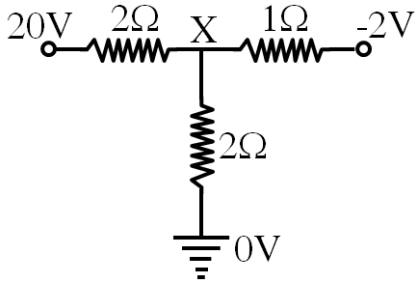
(1) 2995 Ω

(2) 4995 Ω

(3) 10^3 Ω

(4) 10^5 Ω

4) The current in 1Ω resistor is :



(1) 2A

(2) 3A

(3) 4A

(4) 6A

5) In India electricity is supplied at 220V while in USA it is supplied at 110V. For bulbs rated 60W, in both countries. the ratio of the resistances is :

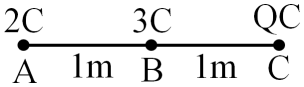
(1) 1 : 2

(2) 2 : 1

(3) 4 : 1

(4) 1 : 1

6) If total potential energy of the system is zero, then find the value of charge 'Q' ?



(1) –3C

(2) $-\frac{3}{2}$C

(3) +4C

(4) +3C

7) Calculate the electric potential at the surface of a gold nucleus. Given, radius of nucleus 6.6 × 10^{-15} m and atomic number of gold is 79.

(1) 1.72 × 10^7 V

(2) 1.72×10^5 V

(3) 2.72×10^7 V

(4) 2.72×10^5 V

8) **Statement I :** Interference and diffraction event is observed only for transverse wave.
Statement II : Polarization phenomenon occurs for longitudinal wave.

(1) Both statement I and statement II are false

(2) Statement I is false but statement II is true

(3) Statement I is true but statement II is false

(4) Both statement I and statement II are true

9) Two coherent sources are 0.3mm apart. They are 0.9 m away from the screen. Wavelength of light used is 2×10^{-6} m. The fringe width will be :-

(1) 9 mm

(2) 2 mm

(3) 4 mm

(4) 6 mm

10) A spherical charged drop has radius r and it's capacitance is C. When such 64 drops combined, then its capacitance will be :-

(1) 4C

(2) 2C

(3) 64 C

(4) 8C

11) A pump can take out 36000 kg of water per hour from a 90 m deep well. If the efficiency of pump is 60%, its power is :

(1) 15 KW

(2) 30 KW

(3) 60 KW

(4) 7.5 KW

12) There is a small hole at bottom in a hollow sphere. The water enters in it when it is taken to a depth of 40 cm in water. The surface tension of water is 0.07 N/m. The diameter of hole is:

(1) 3.5 mm

(2) 0.07 mm

(3) 0.35 mm

(4) 0.7 mm

13) The mass and volume of a body are 4.237 g and 2.5 cm^3, respectively. The density of the material of the body in correct significant figure is :-

(1) 1.6048 g cm^{-3}

(2) 1.69 g cm^{-3}

(3) 1.7 g cm^{-3}

(4) 1.695 g cm^{-3}

14) A ball is dropped from a height of 20 m. If the coefficient of restitution for the collision between ball and floor is 0.5. After second collision with the floor, the ball rebounds to a height.

(1) 10 m

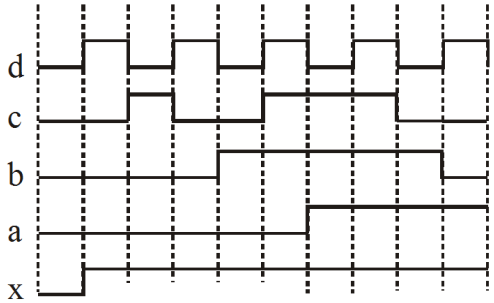
(2) 5 m

(3) 2.5 m

(4) 1.25 m

15)

If a, b, c, d are inputs to a GATE and x is its output, then as per the following time graph, the GATE is :



(1) NAND

(2) NOT

(3) AND

(4) OR

16) The initial mass of a rocket is 2000 kg. Calculate at what rate the fuel should be burnt so that the rocket is given upwards acceleration of 10 m/s^2, from the earth's surface, the gases come out with relative speed of 500 m/s with respect to the rocket is :

(1) 40 kg s^{-1}

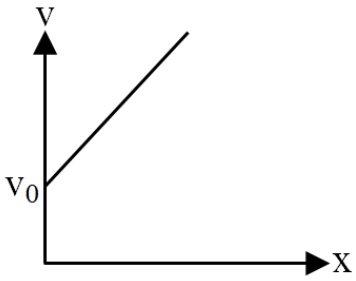
(2) 20 kg s^{-1}

(3) 10 kg s^{-1}

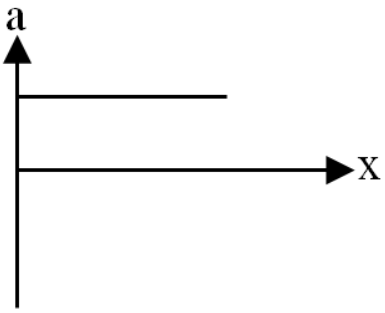
(4) 80 kg s^{-1}

17) The velocity-displacement graph of a particle is shown. Then corresponding acceleration-

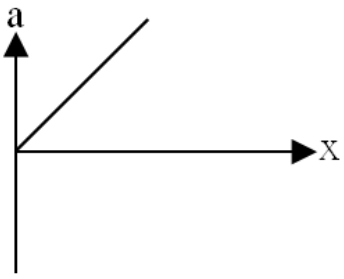
displacement graph will be :



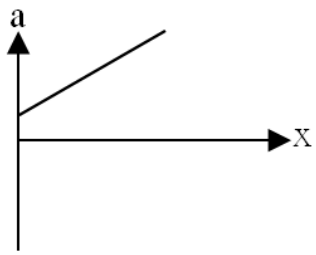
(1)



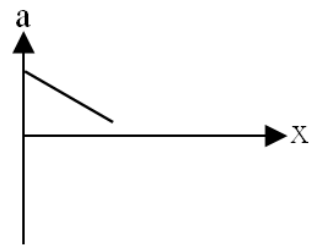
(2)



(3)



(4)



18) A particle accelerates from rest for time t_1 at constant rate 4 m/s^2 and then retards at constant rate 6 m/s^2 for time t_2 and comes to rest. The correct value of $\frac{t_1}{t_2}$ will be :

(1) $\frac{2}{3}$

(2) $\frac{5}{3}$

(3) $\frac{3}{5}$

(4) $\frac{3}{2}$

19) The temperature of a room is 30°C. A body in this room cools from 61°C to 59°C in 4 minutes. The time taken by the body to cool from 51°C to 49°C will be :-

(1) 4 min

(2) 6 min

(3) 5 min

(4) 8 min

20) A nylon rope of 3 cm diameter has a breaking strength equal to 1.5×10^5 N. The breaking strength of a similar rope having 1.5 cm diameter is :

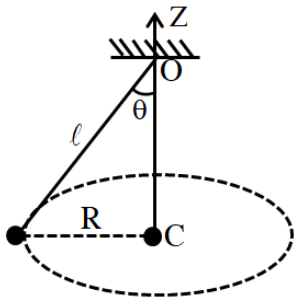
(1) 0.75×10^5 N

(2) 0.375×10^5 N

(3) 3×10^5 N

(4) 6×10^5 N

21) A conical pendulum of length '□' making an angle 'θ' w.r.t z-axis and moves in a circle at the XY plane. The radius of the circle is R and it's centre is vertically below O. The speed of the pendulum, in it's circular path, will be :



(1) $V = \sqrt{Rg \tan\theta}$

(2) $V = \sqrt{\frac{Rg}{\cot\theta}}$

(3) Both 1 & 2

(4) None of these

22) A plane EM-wave propagating x-direction is described by
$E_y = (100\ VM^{-1})\sin(1.5 \times 10^7 t - 0.05x)$.
The peak value of magnetic field of the wave is
($\varepsilon_0 = 8.85 \times 10^{-12}\ C^2m^{-2}N^{-1}$)

(1) 0.33×10^6 T

(2) 0.33×10^{-6} T

(3) 33×10^{-6} T

(4) 3.3×10^{-6} T

23) Current in a 5 mH coil decreases uniformly from 20 A to 10 A in 0.1 second. The average

induced emf will be :-

(1) 0.5 V
(2) 0.2 V
(3) 0.3 V
(4) 0.1 V

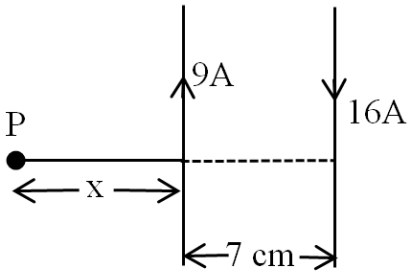
24) In an A.C. circuit the current leads the voltage by $\frac{\pi}{3}$. The components in the circuit are -

(1) R, L
(2) R, C
(3) L
(4) R

25) The wavelength of the photon emitted by an hydrogen atom, when an electron makes a transition from n = 3 to n = 1 state is :

(1) 1810 Å
(2) 1026 Å
(3) 1368 Å
(4) 1540 Å

26) Two long straight conductors are held parallel to each other at 7 cm apart the conductors carry current of 9A & 16A in opposite directions. The distance of Neutral point from the conductor carrying 16A current is :-



(1) 9 cm
(2) 16 cm
(3) 25 cm
(4) $\frac{63}{25}$ cm

27) A long solenoid has 200 turns per cm carries current i and the magnetic field at its centre is 6.28×10^{-2} Wb/m^2. Another long solenoid has 100 turns per cm and it carries current i/2. The value of the magnetic field at its centre is :-

(1) 1.05×10^{-2} Wb/m^2
(2) 1.57×10^{-2} Wb/m^2
(3) 2.56 Wb/m^2

(4) 1.05×10^{-3} Wb/m^2

28) A wire of length 6.28 m is bent into a circular coil of 2 turns If a current of 0.5 A exists in the coil, the magnetic moment of the coil is (Am^2) :-

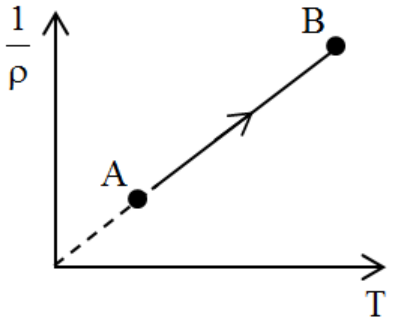
(1) $\frac{\pi}{4}$

(2) $\frac{1}{4}$

(3) π

(4) 4π

29) The figure shows graph between $\frac{1}{\rho}$ and T for an ideal gas, gas undergoes process AB is :- (ρ :



density & T : Temperature)

(1) Isothermal

(2) Isobaric

(3) Isochoric

(4) Adiabatic

30) Temperature of three moles of an diatomic gas changes from T_1 = 100°C to T_2 = 200°C in an isobaric process. If Q, ΔU and ΔW represent heat supplied, change in internal energy and work done respectively; match column -I and column-II (R = $\frac{25}{3}$ J/mole/R)

	Column-I		**Column-II**
(A)	ΔW	(i)	6250 J
(B)	ΔU	(ii)	2.5
(C)	Q	(iii)	2500 J
(D)	$\frac{\Delta U}{\Delta W}$	(iv)	8750 J

(1) A-iii, B-i, C-iv, D-ii

(2) A-i, B-ii, C-iv, D-iii

(3) A-ii, B-i, C-iii, D-iv

(4) A-iv, B-iii, C-ii, D-i

31) Two simple pendulum of length 1 m and 25 m respectively, are initially in phase. After how much

oscillation of smaller length pendulum, they are again in same phase ?

(1) $\frac{4}{3}$

(2) $\frac{5}{4}$

(3) $\frac{7}{5}$

(4) $\frac{8}{7}$

32) Given equation of wave on a string of mass per unit length $\mu = 0.16$ kg/m is $y = 0.02 \sin (x + 30t)$. Find the tension in string :

(1) 12 N

(2) 144 N

(3) 36 N

(4) 100 N

33) Radii of curvature of a converging lens are in the ratio 1 : 2. Its focal length is 6 cm and Refractive index = 1.5, then its radii of curvature are respectively :

(1) 9 cm and 18 cm

(2) 6 cm and 12 cm

(3) 3 cm and 6 cm

(4) 4.5 cm and 9 cm

34) If ${}^{i}u_{j}$ represents refractive index a light ray goes from medium i to medium j, then the product ${}^{2}u_{1} \times {}^{3}u_{2} \times {}^{4}u_{3}$ is equal to :

(1) ${}^{3}u_{1}$

(2) ${}^{3}u_{2}$

(3) ${}^{1}u_{4}$

(4) ${}^{4}u_{1}$

35) A 90 kg body placed at 2R distance from surface of earth experiences gravitational pull of : (R = Radius of earth, g = 10 ms^{-2})

(1) 300 N

(2) 225 N

(3) 120 N

(4) 100 N

36) A satellite of mass m is orbiting the Earth at a height h above its surface. The mass of the Earth is M and its radius R. Match the physical quantities in List I with the expression in List II.

List -I	List - II

(P)	$\frac{GMm}{2(R+h)}$	(1)	Potential energy of the earth satellite system.
(Q)	$\sqrt{\frac{GM}{R+h}}$	(2)	Kinetic energy of the satellite.
(R)	$\frac{-GMm}{R+h}$	(3)	Orbital velocity of the satellite.
(S)	$-\frac{GMm}{2(R+h)}$	(4)	Total energy of satellite.

(1) P → 2 ; Q → 3 ; R → 4 ; S → 1

(2) P → 1 ; Q → 2 ; R → 3 ; S → 4

(3) P → 4 ; Q → 1 ; R → 2 ; S → 3

(4) P → 2 ; Q → 3 ; R → 1 ; S → 4

37) Hydrogen gas absorbs radiations of wavelength λ_0 and consequently emit radiations of 6 different wavelengths of which three wavelengths are shorter than λ_0. Choose the **CORRECT** statement(s).
(a) The final excited state of the atom is n = 4.
(b) The initial state of the atom may be n = 2.
(c) There are two transitions in Balmer series.
(d) There are three transitions belonging to Lyman series.

(1) a and b

(2) a, b and c

(3) a, b, c and d

(4) a, c and d

38) The mass of proton, neutron and helium nucleus are respectively 1.0073 u, 1.0087 u and 4.0015u. The binding energy of helium nucleus is:

(1) 14.2 MeV

(2) 28.4 MeV

(3) 56.8 MeV

(4) 7.1 MeV

39) The de-Broglie wavelength of a proton and α-particle are equal. The ratio of their velocities is :

(1) 4 : 3

(2) 4 : 1

(3) 4 : 2

(4) 1 : 4

40) The least count of vernier callipers is 0.1 mm. The main scale reading before the zero of vernier scale is 10 and the fourth division of the vernier scale coincides with any main scale division. Each main scale division is equal to 1 mm. The measured value should be expressed as :-

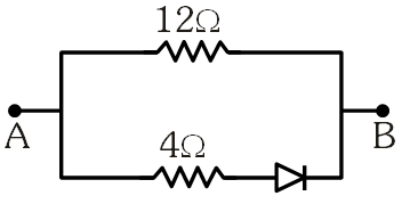
(1) 1.04 mm

(2) 10.4 cm

(3) 1.02 cm

(4) 1.04 cm

41) Find the equivalent resistance of the network shown in the figure between the points A and B, if



(i) $V_A > V_B$ (ii) $V_A < V_B$

(1) 4Ω, 12Ω

(2) 3Ω, 4Ω

(3) 3Ω, 12Ω

(4) 12Ω, 3Ω

42) When a piece of aluminium wire of finite length is drawn to reduce its diameter to half its original value, its resistance will become :-

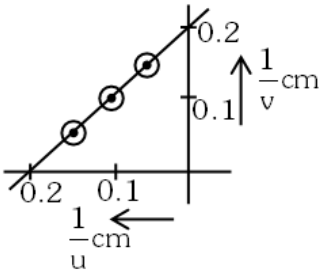
(1) two times

(2) four times

(3) eight times

(4) sixteen times

43) An experiment with convex lens gives certain result which is represented by a student in the shown graph. What would be the power of the lens used ?



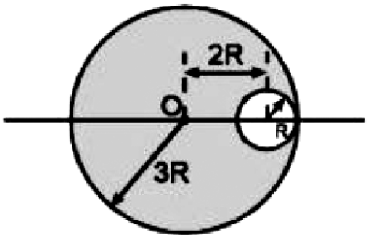
(1) 0.2 D

(2) 1 D

(3) 0.1 D

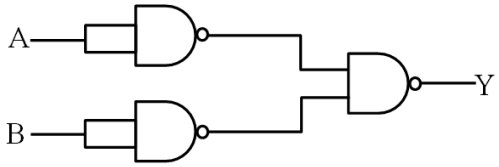
(4) 20 D

44) In the figure shown find out the distance of centre of mass of a system of a uniform circular plate of radius 3 R from point O in which a hole of radius R is cut whose centre is at 2R distance from centre of large circular plate.



(1) R/4
(2) R/5
(3) R/2
(4) none of these

45) The combination of the logic gates shown in the figure below produces :



(1) NOR gate
(2) OR gate
(3) AND gate
(4) XOR gate

BIOLOGY

1) Which type of root form by symbiotic association with cyanobacteria in cycas ?

(1) Tap Root
(2) Fibrous Root
(3) Coralloid Root
(4) Pneumatophore

2) Which of the following is not correct about Lichens ?

(1) Lichens are dual organism containing Phycobiont and Mycobiont
(2) They can't be grow in presence of polluted air
(3) They grow very fast
(4) They often grow in most inhospitable places such as barren rock.

3) The members of kingdom have cellulose in their cell wall are :

(1) Monera, Protista and Plantae only
(2) Protista, Fungi and Plantae only
(3) Plantae and Protista
(4) Plantae only

4) _____ are members of solanaceac family.

(1) Petunia, Tulip
(2) Belladonna, Aloe
(3) Tomato, Brinjal
(4) Tobacco, Lupin

5) Due to formation of false septum ovary become bilocular in -

(1) Mustard
(2) Argemone
(3) China rose
(4) Both (1) & (2)

6) Which of the following is first stable product in C4 cycle ?

(1) Oxaloacetic acid
(2) RuBP
(3) Phosphoenol pyruvic acid
(4) 3-phosphoglyceric acid

7) ATP synthesis is linked with _____.

(1) Development of a proton gradient across a membrane.
(2) Development of a electron gradient across a membrane.
(3) Excitation of electrons
(4) Development of Redox potential

8) Match List I with List II :

	List I		**List II**
A	Carboxylation products	i	Required one ATP to form RuBP
B	Reduction products	ii	RuBP
C	Regeneration	iii	2 mol of 3-PGA
D	CO_2 acceptor	iv	Triose phosphate

Choose the correct answer from option given below :
(1) A-iii, B-iv, C-i, D-ii
(2) A-ii, B-iv, C-iii, D-i
(3) A-iv, B-iii, C-i, D-ii
(4) A-i, B-ii, C-iii, D-iv

9) Read the following statements.
Statement – I : Carbon dioxide is the major limiting factor for photosynthesis.
Statement – II : Current availability of CO_2 level is not limiting to the C_3 plants.
Choose the correct answer from the option given below.

(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Only statement I is correct
(4) Only statement II is correct

10) Fill in the blanks with right option :
Calvin pathway occurs in _____.

(1) C_3 plants only

(2) C_3 & C_4 plants

(3) C_4 plants only

(4) CAM plants only

11) Which of the following characterstic is not shown by C_4 plants ?

(1) Kranz anatomy

(2) Can tolerate higher temperature

(3) Response to high light intensities

(4) Photorespiration.

12) Other thylakoid pigment like chlorophyll-b, Xanthophylls and catotenoids which are called __A__. identify A ?

(1) A- chief pigment

(2) A-photosynthesis

(3) A-accessory pigment

(4) A-normal pigment

13) When pericarp is thick and fleshy then it is differentiated into the outer ___A___, the middle ___B___ and the inner ___C___.
Find out the (A), (B) and (C)

Options :	**A**	**B**	**C**
(1)	Endocarp	Mesocarp	Epicarp
(2)	Epicarp	Endocarp	Mesocarp
(3)	Mesocarp	Epicarp	Endocarp
(4)	Epicarp	Mesocarp	Endocarp

(1) 1

(2) 2

(3) 3

(4) 4

14) **Statement-I :** Gynoecium is bicarpellary obliquely placed ovary superior, bilocular placenta swollen with many ovules, axile placentation in liliaceae family.
Statement-II : Underground modification of stem like bulbs, corms and rhizomes are found in members of liliaceae.

(1) Both statements I and II are incorrect

(2) Statement I is correct but statement II is incorrect

(3) Statement I is incorrect but statement II is correct

(4) Both statements I and II are correct

15) **Statement-I :** Bacteria may be motile or non motile.
Statement-II : Motility in bacteria depend upon presence of pili.

(1) Statement I is correct.

(2) Statement II is correct.

(3) Statement I & II incorrect.

(4) Both statement are correct.

16) Match the column-I with column-II and choose the correct option :

Column-I		**Column-II**	
(i)	Monoecious plant	(a)	Papaya
(ii)	Dioecious plant	(b)	*Hydrilla*
(iii)	Cleistogamous flower	(c)	Castor
(iv)	Water pollination	(d)	*Commelina*

(1) (i) b, (ii) a, (iii) d, (iv) c

(2) (i) c, (ii) a, (iii) d, (iv) b

(3) (i) c, (ii) d, (iii) a, (iv) b

(4) (i) c, (ii) a, (iii) b, (iv) d

17) Which of the following structure exhibits a fascinating array of patterns and design?

(1) Intine

(2) Exine

(3) Germ pore

(4) Filiform apparatus

18) Select incorrect statement ?

(1) Gynoecium represents the female reproductive part of the flower.

(2) Pollen grains can be rich in nutrients

(3) Megasporangium is also known as pollen sac

(4) Chalaza representing the basal part of the ovule

19) **Assertion (A) :** The cells of the tapetum are initially diploid and uninueleate but they become polyploid and multinucleate.
Reason (R) : Tapetal cells undergo endomitosis and free nuclear division, respectively.

(1) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**.

(2) **(A)** is correct but **(R)** incorrect.

(3) **(A)** is incorrect but **(R)** correct.

(4) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**.

20) Read the following statements and choose the correct option.
Statement-I : Geitonogamy is functionally cross pollination involving pollinating agent, genetically it is similar to autogamy since the pollen grains come from the same plant.
Statement-II : The correct sequence of embryo development in dicots is-
Zygote → globular embryo → proembryo → heart shaped embryo → mature embryo.

(1) Both statements are correct.

(2) Both statements are incorrect.

(3) Statement-I is correct but Statement-II is incorrect.

(4) Statement-I is incorrect but Statement-II is correct.

21) Given below are two statements one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : Walter Sutton and Theodore Boveri noted that the behaviour of chromosomes was parallel to the behaviour of genes
Reason (R) : Experimental verification of the chromosomal theory of inheritance was done by Thomas Hunt Morgan.

(1) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

(2) Both **(A)** and **(R)** are correct and **(R)** is not the correct explanation of **(A)**

(3) **(A)** is are correct but **(R)** is not correct

(4) **(A)** is not correct but **(R)** is correct

22) Identify the correct statement :-
(A) Dominance is an autonomous feature of gene.
(B) In co-dominance the F_1 generation can resembles both parents.
(C) Size of starch grain in pea plant is an example of incomplete dominance.
(D) In pleiotropy many genes exhibit single phenotype.
Choose the correct answer from the options given below :

(1) A and D only

(2) B and C only

(3) A, B, C and D

(4) B, C and D

23) **Statement-I :** The segregation of alleles is a random process and so there is a 50% change of a gamete containing either alleles.
Statement-II : Non disjunction may be the exception of segregation.

(1) Both statement I and II are correct

(2) Both statement I and II are incorrect

(3) Statement I is correct and statement II is incorrect

(4) Statement I is incorrect and statement II is correct

24) Given below are two statements :
Statement I : The term recombination to describe the generation of non-parental gene

combinations.
Statement II : It was also observed that in a large number of insects the mechanisms of sex determination is of the XO type.
Choose the correct answer from the options given below :

(1) Both Statement-I and Statement-II are correct.

(2) Both Statement-I and Statement-II are incorrect.

(3) Statement-I is correct and Statement-II are incorrect.

(4) Statement-I is incorrect and Statement-II are correct.

25) The law of dominance is used to explain the expression of only one of the parental characters in a monohybrid cross in the F_1 and the expression of both in the F_2. It also explains the proportion of _____ obtained at the F_2.

(1) 1 : 2 : 1

(2) 3 : 1

(3) 1 : 1

(4) 9 : 3 : 3 : 1

26) In the F_1 RrYy plant, four genotypes of gametes are __a__ each with a frequency of __b__ percent of the total gametes produced.
Find the correct fill in blanks respectively.

(1) a-RY, Rr, Yy, yr; b-50

(2) a-RR, YY, rr,yy; b - 25

(3) a-Ry, Yy, Rr, yr; b - 50

(4) a-RY, Ry, rY, ry; b-25

27) Select the incorrect option w.r.t. organisms and their set of chromosomes.

	Organism	**Male**	**Female**
(1)	Human	2A + XY	2A + XX
(2)	Birds	2A + ZW	2A + ZZ
(3)	Grasshopper	2A + XO	2A + XX
(4)	*Drosophila*	2A + XY	2A + XX

(1) 1

(2) 2

(3) 3

(4) 4

28) **Statement-I :** ABO blood groups are controlled by gene I.
Statement-II : The alleles I^A & I^B produce slightly different form of sugar.

(1) Both Statement-I and Statement-II are incorrect.

(2) Statement-I is correct and Statement-II are incorrect.

(3) Statement-I is incorrect and Statement-II are correct.

(4) Both Statement-I and Statement-II are correct.

29) Which of the following RNA polymerase enzyme transcribes hn-RNA :

(1) RNA polymerase-I

(2) RNA polymerase-II

(3) RNA polymerase-III

(4) RNA polymerase I and II

30) **Statement-I :** Viruses having RNA genome show shorter life span and evolve faster.
Statement-II : The unequivocal proof that DNA is genetic material came from the experiments of Hershey and Chase.

(1) Both statement I and II are correct

(2) Both statement I and II are incorrect

(3) Statement I is correct and statement II is incorrect

(4) Statement I is incorrect and statement II is correct

31) Find the incorrect statement :

(1) DNA replication takes place in S-phase of the cell cycle of eukaryotes.

(2) Semi-conservative method of DNA replication was first proposed by Watson & Crick.

(3) DNA as an acidic substance present in nucleus was first identified by Franklins.

(4) DNA is a long polymer deoxyribonucleotides.

32) Given below are two statements :
Statement-I : The two polynucleotide chains in DNA are parallel.
Statement-II : The okazaki fragments in DNA chain are polymerised in the 5'-to-3' direction.
Choose the correct answer from the options given below.

(1) Both Statement-I and Statement-II are incorrect.

(2) Statement-I is correct and Statement-II are incorrect.

(3) Statement-I is incorrect and Statement-II are correct.

(4) Both Statement-I and Statement-II are correct.

33) Which of the following statement is true :-
(A) One turn of the helix in a B-form DNA is approximately 34 nm.
(B) A typical nucleosome contains 200 bps of DNA helix.
(C) In *E.coli* the total length of DNA is 1.36 μm.

(1) B & C only

(2) A & B only

(3) B only

(4) A,B and C

34) Find the correct statement.

(1) m-RNA is called adapter molecule.

(2) In capping, methyl guanosine triphosphate is added to the 3' end of hnRNA.

(3) Cistron is a segment of DNA coding for a polypeptide.

(4) Aminoacylation of t-RNA is the last phase of translation

35) Find the incorrect statement.

(1) The formation of DNA from RNA is known as reverse transcription.

(2) Single amino acid coded by more than one codon is called 'Degeneracy of genetic code'.

(3) Sigma factor (σ) recognises the terminator site of DNA.

(4) Transcription proceeds in 5' → 3' direction.

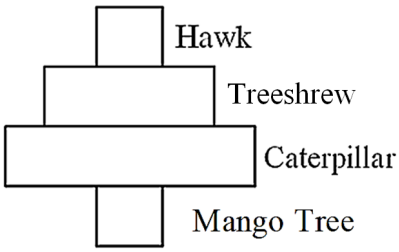
36) **Statement-I :** The approach of human genome project which focused on identifying all the genes that are expressed as RNA is referred as sequence annotation.
Statement-II : 99.9 percent of base sequence among human is the same.

(1) Both Statement-I and Statement-II are correct.

(2) Statement-I is correct and Statement-II are incorrect.

(3) Statement-I is incorrect and Statement-II are correct.

(4) Both Statement-I and Statement-II are incorrect.

37) The diagram shows a pyramid of number in a food chain - 

What type of organism are caterpillars ?

(1) Primary producer

(2) Primary consumer

(3) Quaternary consumer

(4) Secondary consumer

38) Below two statement are given
Statement - I : Vertical distribution of different species occupying different levels is called stratification.
Statement - II : Productivity is expressed in term of gm^{-2} yr^{-1}.

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct.

(4) Both **Statement I** and **Statement II** are correct.

39) Humus is important for plant growth because :

(1) It is partially decomposed

(2) It is derived from only leaves

(3) It is rich in nutrients & increase the water holding capacity of soil.

(4) It is made up of living organic matter

40) Hermit crab is the mode of transportation to the Sea-anemone and provide defense system to the Hermit crab, is are example of ?

(1) Co-evolution

(2) Predation

(3) Amensalism

(4) Commensalism

41) **Assertion (A)** :- Stratification is based on mode of arrangement of various growth forms of plants.
Reason (R) :- There are different types of population (species) found in the community this is called species diversity.
Choose the correct option :

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

42) Given below are two statements : One is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A) : The cattle egret bird and grazing cattle in close association. It is classic example of mutualism.
Reason (R) : Plants generally need the help of animals for pollinating their flowers and dispersing their seeds.
In the light of the above statements, choose the correct answer from the options given below :

(1) Both A and R are true and R is correct explanation of A.

(2) Both A and R are true but R is not correct explanation of A.

(3) A is true but R is false.

(4) A is false but R is true.

43) Below two statement are given :
Statement-I : Some organisms breed only once in their life time while others breed many times during their life time.
Statement-II : Some organisms produce a large number of small-sized offspring while others produce a small number of large-sized offspring.

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct.

(4) Both **Statement I** and **Statement II** are correct

44) Given below are two statements.
Statement I :- An individual may have births and deaths, but a population has birth rates and death rates.
Statement II :- Population ecology links ecology to population genetics and evolution.
In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Both **Statement I** and **Statement II** are incorrect.
(2) **Statement I** is correct but **Statement II** is incorrect.
(3) **Statement I** is incorrect but **Statement II** is correct
(4) Both **Statement I** and **Statement II** are correct.

45) Which one of the following statement is incorrect?

(1) Strict protection of Biodiversity hotspots could reduce ongoing mass extinctions by almost 30%.
(2) 20% of the total oxygen in the earth's atmosphere comes form Amazon rain forest.
(3) The Earth Summit was held in Johannesburg, South Africa on sustainable development.
(4) More than 25% of drugs currently sold in the market world wide are derived from plants.

46) Virus infected cells secrete a protein called __A__ which protect __B__ cells.

(1) A → Interleukins, B → non-infected
(2) A → Interferon, B → infected
(3) A → Interferon, B → non-infected
(4) A → Lymphokines, B → Lymphocytes

47) _____ is the tendency of the body to manifest a characteristic and unpleasant withdrawal syndrome if regular dose of drugs/alcohol is abruptly discontinued.

(1) Addiction
(2) Adolscence
(3) Dependence
(4) Adaptation

48) **Assertion :** Symptoms of amoebiasis include cramps stool with excess mucous & blood clots.
Reason : Housefly act as mechanical carriers and serve to transmit parasite from faeces to food.

(1) Both Assertion and Reason are True and Reason is the correct explanation of Assertion.
(2) Both Assertion and Reason is True but Reason is not the correct explanation of Assertion.
(3) Assertion is True but Reason is False.
(4) Both Assertion and Reason are False.

49) All of the following evolved from sauropsids except-

(1) Snakes
(2) Turtles
(3) Tuataras

(4) Mammals

50) **Statement-I :** The process of evolution of different species in a given geographical area starting from a point and literally radiating to other areas of geography (habitats) is called convergent evolution.
Statement-II : Darwin's finches represent one of the best example of convergent evolution.

(1) Both the statements are correct
(2) Both the statements are incorrect
(3) Statement II is correct, but statement I is incorrect
(4) Statement II is incorrect, but statement I is correct

51)

Identify the mismatched pair :-

(1) Labeo - Rohu
(2) Trygon - Catla
(3) Torpedo - Electric organ
(4) Scoliodon - Dog fish

52) Select the incorrect statement for molluscs :

(1) Body divided into head, foot and visceral hump
(2) A mantle layer present over calcareous shell
(3) Feather like gills present in mantle cavity
(4) File-like rasping organ radula present

53) Stomochord is present in :-

(1) Proboscis region
(2) Collar region
(3) Trunk region
(4) Proboscis gland

54) The partial pressure of oxygen in the alveoli of the lungs is-

(1) Less than that in the blood
(2) Less than that of CO_2 in the blood.
(3) Equal to that in the blood
(4) More than that in the blood

55) Deficiency of vasopressin primarily results in :-

(1) Excessive secretion of urochrome.
(2) Increased volume of urine.
(3) Decreased volume of urine.

(4) Change in pH from acidic to alkaline range.

56) Identify the pair of hormones which generate secondary messenger to perform their hormonal action?

(1) Cortisol, Progesterone
(2) Aldosterone, Oestrogen
(3) FSH, Insulin
(4) Progesterone, Testosterone

57) Ciliated epithelium are mainly present in the inner surface of hollow organs like _____?

(1) Bronchioles
(2) Fallopian tube
(3) Trachea
(4) All the above

58) Which of the technique is not a part of in Vivo fertilisation :-

(1) G.I.F.T
(2) A.I
(3) Z.I.F.T
(4) Both (1) & (2)

59) Match the column-

Column-I		**Column-II**	
(i)	Fibrinogen	(A)	Involve in defense
(ii)	Globulin	(B)	Help in clotting
(iii)	Albumin	(C)	Help in osmotic balance

Which of the following is correctly match ?
(1) i–A, ii–B, iii–C
(2) i–C, ii–B, iii–A
(3) i–B, ii–A, iii–C
(4) i–B, ii–C, iii–A

60) During which phase of the cardiac cycle do both atria and ventricles Relax ?

(1) Systole
(2) Joint diastole
(3) Contraction
(4) Both (1) and (3)

61) The secretion of __(a)__ help the __(b)__ enter into the cytoplasm of the __(c)__ through __(d)__and plasma membrane.

Find the correct option which fills above blanks correctly.

(1)
(a) Acrosome
(b) Ovum
(c) Sperm
(d) Zona pellucida

(2)
(a) Mitochondria
(b) Sperm
(c) Ovum
(d) Zona pellucida

(3)
(a) Acrosome
(b) Sperm
(c) Ovum
(d) Zona pellucida

(4)
(a) Mitochondria
(b) Sperm
(c) Ovum
(d) Zona pellucida

62) Select the incorrect statement :-

(1) The male sex accessory ducts include rete testis, vasa efferentia, epididymis and vas deferens.

(2) Epididymis is located along the anterior surface of each testis.

(3) Vas deferens receives a duct from seminal vesicle to form ejaculatory duct.

(4) Male sex accessary ducts store and transport the sperms from the testis to the outside through urethra.

63) During pregnancy level of hormones increased several folds in maternal blood, these essential for :

(1) Fetal growth

(2) Metabolic changes in mother

(3) Maintenance of pregnancy

(4) All of the above

64) The ovum released by the ovary is transported to which region of fallopian tube where fertilisation takes place :-

(1) Isthmus

(2) Ampulla

(3) Infundibulum

(4) Cervix

65) **Statement-I :-** Axolemma is impermeable to negative charged proteins present in the axoplasm.
Statement-II :- Resting membrane maintained by active-transport of ions by Na^+-K^+ pump which transports $3Na^+$ outwards for $2K^+$ into the cell.

(1) Both Statement-I and Statement-II are correct.

(2) Both Statement-I and Statement-II are incorrect.

(3) Statement-I is correct but Statement-II is incorrect.
(4) Statement-I is incorrect but Statement-II is correct.

66) The large regions of cerebral cortex that are neither clearly sensory nor motor in function known as

(1) Sensory areas
(2) Motor areas
(3) Association areas
(4) Limbic system

67) Muscles which have characteristic striations & involuntary in nature are

(1) Muscles of the eyelids
(2) Muscles assisting locomotion
(3) Muscles of the heart
(4) Muscles in the wall of alimentary canal

68) Bones of the limbs along with their girdles constitutes the _____ :–

(1) Axial skeleton
(2) Pectoral skeleton
(3) Rib cage
(4) Appendicular skeleton

69) DNA replication occurs at :-

(1) G_0 & G_1
(2) G_2-stage
(3) S-stage
(4) Metaphase

70) Which of the following is not observed during metaphase?

(1) Position of Centromere
(2) Number of Chromosome
(3) Nucleolus
(4) Size of Chromosome

71) According to fluid mosaic model of cell membrane, the quasi - fluid nature of lipid -

(1) inhibits lateral movement of protein with in overall bilayer.
(2) Describes the arrangements of DNA in the membrane.
(3) Describes the arrangement of ribosome in the membrane.
(4) Enables lateral movement of proteins with in the overall bilayer

72) In the polysaccharide the individual monosaccharide are linked by :-

(1) Glycosidic bond
(2) Phosphodiester bond
(3) Hydrogen bond
(4) Peptide bond

73) **Assertion :** If one wants to recover many copies of the target DNA , it should be cloned in a vector whose origin support high copy number.
Reason : Origin of replication (ori) is responsible for controlling the copy number of the linked DNA.

(1) Both assertion & reason are true & the reason is a correct explanation of the assertion.
(2) Both assertion & reason are true but reason is not a correct explanation of the assertion.
(3) Assertion is true but the reason is false.
(4) Both assertion & reason are false.

74) The process by which restriction endonuclease cut the DNA molecule :-

(1) Reduction
(2) Oxidation
(3) Hydrolysis
(4) Carboxylation

75)

Given below are two statements :
Statement I : The Indian Government has set up GEAC which will make decisions regarding the validity of GM research.
Statement II : Biopiracy is the term used to refer to the use of bio-resources by indigenous or local people.
In the light of the above statements, choose the correct answer from the options given below :

(1) **Statement** I is true but **Statement II** is false
(2) **Statement I** is false but **Statement II** is true
(3) Both **Statement I** and **Statement II** are true
(4) Both **Statement I** and **Statement II** are false

76) Golden rice is a genetically modified crop plant where the incorporated gene is meant for biosynthesis of :-

(1) Vitamin 'C'
(2) Omega 3
(3) Vitamin 'A'
(4) Vitamin 'B'

77) Arrange the following events of meiosis in ascending order of their occurence & select the **correct** option from given below options :

A. Disjunction of homologous chromosomes
B. Centromere division
C. Terminalization of chiasmata
D. Dissolution of synaptonemal complex
E. Formation of bivalent
F. Activity of recombinase enzyme
Options :-

(1) E → F → D → C → B → A
(2) E → F → D → C → A → B
(3) F → E → B → C → A → B
(4) F → E → B → C → B → A

78) Structure with an internal 9 + 0 morphology is :-

(1) Cilia
(2) Centriole
(3) Basal body
(4) Both (2) and (3)

79) Which one of the following is present in minimum amount in human body ?

(1) Hydrogen
(2) Oxygen
(3) Nitrogen
(4) Silicon

80) The first letter of the name of restriction endonuclease came from the :

(1) genus of organism
(2) species of organism
(3) family of organism
(4) class of organism

81) The two polypeptides of human insulin are linked together by :-

(1) Hydrogen bonds
(2) Phosphodiester bond
(3) Covalent bond
(4) Disulphide bridges

82)

Which of the following is a matching set of a phylum and its three example?

(1) Porifera : Spongilla, Euplectella, Pennatula
(2) CNIDARIA : Ctenoplana, Physalia, Aurelia

(3) Platyhelminthes : Planaria, Schistosoma, Enterobius

(4) Mollusca : Loligo, Pinctada, Octopus

83) In cockroach exchange of gases take place at the _____ by diffusion.

(1) Seiracles

(2) Trachea

(3) Tracheoles

(4) Haemocoel

84) A number of marsupials, each different from the other evolved from an ancestral stock but all within the _____ continent.

(1) African

(2) Indian

(3) Australian

(4) None of these

85) **Assertion :** Production of thymosins is decreased in old individuals.
Reason : Thymus is degenerated in old individual.

(1) Assertion and reason is correct and reason is correct explanation of assertion.

(2) Assertion and reason is correct but reason is not correct explanation of assertion.

(3) Assertion is correct but reason is false.

(4) Both assertion and reason is false.

86) Addison's disease is caused due to underproduction of hormones by -

(1) Adrenal medulla

(2) Adrenal cortex

(3) Anterior pituitary

(4) Thyroid gland

87) Which of the following method is non hormonal ?

(1) LNG-20

(2) Oral contraceptive

(3) Multiload 375

(4) Lactational amenorrhea.

88) Given below are two statements :
Statement-I : Secretions of epididymis, vas deferens, seminal vesicle and prostate are essential for maturation & motility of sperm.
Statement-II : The function of male sex accessory ducts and glands are maintained by FSH.
In the light of the above statements, choose the correct answer from the options given below :

(1) Both statement-I and statement-II are false.

(2) Statement-I is correct but statement II is false

(3) Statement-I is incorrect but statement-II is true.

(4) Both statement-I and statement-II are true.

89) During inspiration :-

(1) Atmospheric air is drown out

(2) Intrapulmonary pressure decreases

(3) Pulmonary volume decrease

(4) Intrapulmonary pressure increases

90) ___(i)___ is absent or highly reduced in cortical nephrons.
Fill the blanks (i)

(1) Vasa recta

(2) Collecting duct

(3) Loop of henle

(4) PCT

ANSWER KEYS

CHEMISTRY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	4	3	3	1	4	3	3	4	1	4	3	3	3	2	4	1	1	3	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	2	4	3	1	1	3	3	4	4	1	2	3	3	3	1	4	1	2	4
Q.	41	42	43	44	45															
A.	3	4	3	2	3															

PHYSICS

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	4	2	2	4	3	2	1	1	4	1	1	2	3	4	4	4	3	4	2	2
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	3	2	1	2	2	2	2	1	2	1	2	2	4	4	4	4	3	2	2	4
Q.	86	87	88	89	90															
A.	3	4	4	1	2															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	3	3	3	3	4	1	1	1	3	2	4	3	4	3	1	2	2	3	1	3
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	2	2	1	1	2	4	2	4	2	1	3	3	3	3	3	3	2	4	3	4
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	1	4	4	4	3	3	3	2	4	2	2	2	2	4	2	3	4	3	3	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	2	4	2	1	3	3	4	3	3	4	1	1	3	1	3	2	4	4	1
Q.	171	172	173	174	175	176	177	178	179	180										
A.	4	4	3	3	1	2	3	2	2	1										

SOLUTIONS

CHEMISTRY

1)

Acidic Nature ∝ +ve O.S. of Non-Metal

2)

Due to lanthanoid contraction size of 4d $\simeq$ 5d series elements

3)

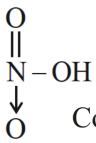
Iodine can show oxidation state from –1 to +7

4) B_2 = Total el^- = 10
$\sigma_{1s}^2\sigma_{1s}^{*2}\sigma_{2s}^2\sigma_{2s}^{*2}$ $[\pi_2 p_x^1 = \pi_2 pp^1]$
B.O = 1 & Paramagnetic

5) Ionic character $\propto \frac{1}{\text{Polarising power of cation}}$
As positive oxidation state increases or, size of cation decreases, polarizing power of cation increases hence, the correct order of ionic character is :
MCl > MCl_2 > MCl_3

6) NCERT XIth Pg # 104, Table 4.2

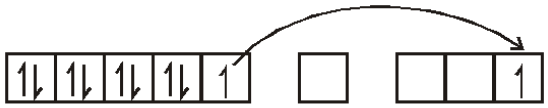
7) $B.E. \propto \frac{1}{\text{no. of L.P.}}$
NCERT XI Pg. # 108 (Part-I)

8)  Covalency = 4

9) μ = NH_3 > NF_3 > BF_3
μ = 1.47 D μ = 0.24 D μ = 0

10) Ionisation Isomerism

11) $[Cu(NH_3)_4]^{+2}$
$Cu^{+2} = 4s^0\ 3d^9$



dsp^2 , paramagnetic

12)

IUPAC
$[Ag(H_2O)_2]\ [Ag(CN)_2]$
Coordination number = 2,
Oxidation state = Ag^{+1}
Diaquasilver(I) dicyanidoargentate(I)

13)

As per surface area.

14)

$NF_3 + H_2O \rightarrow$ No reaction

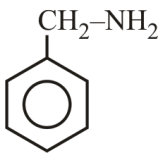
15) $CrO_4^{2-} \underset{OH^-}{\overset{H^+}{\rightleftharpoons}} Cr_2O_7^{2-}$

16) NCERT-XII, Part-II ; Page-321, 322

17)

NCERT XII, Pg. # 271, Part - 2

18)

Duo to localized lone pair 

is the strongest base

20)

Fact

22)

Fact

26)

Fact

28) Theory based NCERT page no. 349

31) $\because 2\pi r = 2\lambda \quad 2\pi r = 2x \quad r = \frac{x}{\pi}$

32) $\Delta x \cdot m\Delta V = \frac{h}{4\pi}$

$\Rightarrow \Delta V = \frac{h}{4\pi m(\Delta x)}$

$= \Delta V = 0.57 \times 10^{7}\ ms^{-1}$

33)

By LCP,
If pressure $\uparrow$, Eqm will shift

where volume is less $V_{ice} > V_{water}$ ⇒Forward direction will be forward.

34) $N_2O_4 \rightleftharpoons 2NO_2$

1 0

$1 - \alpha$ 2α

$$K_p = \frac{(n_{NO_2})^2}{(n_{N_2O_4})} \times \left[\frac{P_1}{\Sigma n}\right]^1$$

For 33% dissociation:

$$K_p = \frac{(2 \times 33)^2}{67} \times \left[\frac{P_1}{133}\right]^1$$

For 50% dissociation:

$$K_p = \frac{(2 \times 50)^2}{50} \times \left[\frac{P_2}{150}\right]^1$$

$$\therefore \frac{P_1}{P_2} = \frac{133 \times 67 \times (2 \times 50)^2}{(2 \times 33)^2 \times 150 \times 50} = \frac{8}{3}$$

35) $BaF_2 \rightleftharpoons Ba^{+2} + 2F^-$

$K_{sp} = [Ba^{+2}]\ (0.1)^2$

$[Ba^{+2}] = \frac{10^{-6}}{(0.1)^2}$

$[Ba^{+2}] = 10^{-4}\ M$

36) $CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$

0 millimoles 5 millimoles 0 0

↓ ↓

5 millimoles 0 5 millimoles

so final solution is buffer having (salt) = (acid)

$pH = Pk_a + \log \frac{[Salt]}{[Acid]}$

since (Salt) = (Acid)

⇒ $pH = Pk_a = 4.7$

37) $\Delta S = 4 \times \log\left(\frac{20}{2}\right) \times 2.3 = 9.2$

38) In adiabatic expansion temperature decrease and isothermal expansion temperature is constant, area under the curve ∝ work.

39) $Fe(OH)_3$ is lyophobic sol so can be prepared by special method.

40) $\Delta G_3^0 = \Delta G_2^0 - -\Delta G_1^0 \; E_3^0 = 3E_2^0 - -2E_1^0$

= 0.772

41) $\frac{W_1}{E_1} = \frac{W_2}{E_2} \Rightarrow \frac{W_1}{W_2} = \frac{108}{31.5}$

42) $K = \frac{2.303}{t} \log\left(\frac{(n-1)P^o}{nP^o - P_t}\right)$

43) $K = \frac{2.303}{t} \log\left(\frac{a}{a-x}\right)$

$K = \frac{2.303}{32} \log\left(\frac{100}{1}\right)$

$t = \frac{2.303 \times 16}{2.303} \log\left(\frac{100}{0.1}\right)$ = 48 min

44)

The one which shows negative deviation from raoults law form maximum boiling azeotropes.

45)

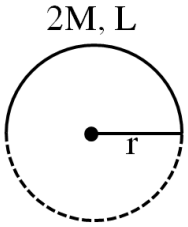
$P_S = P_A^o X_A + P_B^o (1 - X_A)$ $X_B = 0$

If $X_A = 0$ then $P_S = P_A^o (1 - X_B) + P_B^o X_B$

$P_S = P_B^o$ $P_S = P_A^o$

$P = 254 - 119\ X_A$
$P = 254 - 119\ X_0$
$P = P_B^o = 254$
$P = 254 - 119\ (1 - X_B)$
$P = P_A^o = 254 - 119 = 135$

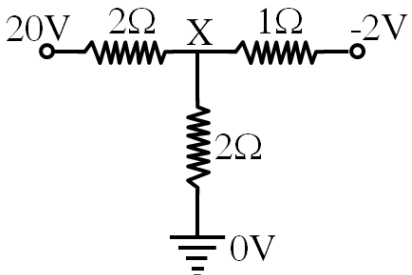
PHYSICS



46)
$\pi r = L$
$\Rightarrow r = \frac{L}{\pi}$
$\Rightarrow I = \frac{2ML^2}{\pi^2}$

47) $I_{Remaing} = I_{Large} - I_{small}$
$$I_{Remaing} = \frac{(9m)R^2}{2} - \left\{ \frac{m'}{2}\left(\frac{R}{3}\right)^2 + m'\left(\frac{2R}{3}\right)^2 \right\}$$
$\Rightarrow m' = m$
$\Rightarrow I_{Remaining} = 4mR^2$

48) G = galvanometer resistance $= \frac{\text{current sensitivity}}{\text{voltage sensitivity}} = \frac{5}{1} = 5\Omega$
Full scale deflection current $i_g = 100 \times \frac{1mA}{5} = 20mA$
To convert in to voltmeter of range 1V×100 = 100 V
$$R = \frac{V}{I_g} - G \Rightarrow R = \frac{100}{20 \times 10^{-3}} - 5 = 4995\Omega$$
.



49) Using KCL of junction X :-
$$\frac{X-20}{2} + \frac{X-0}{2} + \frac{X+2}{1} = 0$$
$X - 20 + X + 2X + 4 = 0$
$4X = 16 \Rightarrow X = 4V$

$I_{1\Omega} = \frac{4-(-2)}{1} = 6A$.

50) $P = \frac{V_1^2}{R_1} = \frac{V_2^2}{R_1} \Rightarrow \frac{R_1}{R_2} = \left(\frac{V_1}{V_2}\right)^2 = \left(\frac{220}{110}\right)^2 = \frac{4}{1}$.

51) O = $\frac{6k}{1} + \frac{3kQ}{1} + \frac{2kQ}{2}$

Q = $-\frac{3}{2}C$

52) q = Ze

= $79 \times 1.6 \times 10^{-19}$ C

V = $\frac{kq}{r}$

= $\frac{9 \times 10^9 \times 1.26 \times 10^{-17}}{6.6 \times 10^{-15}}$

= 1.72×10^7 V

53) Interference and diffraction can be observed for both transverse and longitudinal waves, however polarization can occur only in transverse waves.

54) $\beta = \frac{\lambda D}{d}$

= $\frac{2 \times 10^{-6} \times 0.9}{0.3 \times 10^{-3}}$

= 6×10^{-3} m

β = 6 mm

55) capacitance of small drop C = $4\pi\varepsilon_0 r$

64 such drops combine to form 1 large drop.

 Volume of large drop = 64 × volume of small drop

$\frac{4}{3}\pi R^3 = 64 \times \frac{4}{3}\pi r^3$

R = 4r

 capacitance of big drop

$C_{big} = 4\pi\varepsilon_0 R = 4\pi\varepsilon_0 4r$

= 4C

Formula method :

$C = n^{1/3} C_0 = 64^{1/3} C_0$

= 4 C_0

56) efficiency = $\frac{P_0}{P_i} \times 100$

$60 = \frac{(mgh/time)}{P_i} \times 100$

$P_i = \frac{100}{60}\frac{mgh}{3600} = \frac{5}{3} \times \frac{36000 \times 10 \times 90}{3600}$

= 15 KW

57) $\frac{2T}{r} = h\rho g$

$r = \frac{2T}{\rho gh} = \frac{2 \times 0.07}{10^3 \times 10 \times 0.4} = 0.035$ mm

D = 2r = 0.07 mm

58) $\rho = \frac{m}{v} = \frac{4.237g}{2.5cm^3} = 1.6948$ g/cm^3

rounding off the number = 1.7 g/cm^3

59) $H_n = e^{2n}H_0$

$= e^4 n_0 = \left(\frac{1}{2}\right)^4 \times 20 = \frac{20}{16} = \frac{5}{4} = 1.25m$

60)

Output of OR gate is 0 when all inputs are 0 & output is 1 when atleast one of the input is 1.
Observing output x :- It is 0 when all inputs are 0 & it is 1 when atleast one of the inputs is 1.
 OR gate

61) $F_T - mg = ma$

$\frac{vdm}{dt} - mg = ma$

$\frac{dm}{dt}\frac{m(g+a)}{v} = \frac{2000(10+10)}{500} = 80kgs^{-1}$

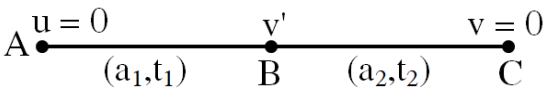
62) $v = v_0 + mx$

$\frac{dv}{dx} = m$

acceleration = $v\frac{dv}{dx} = (v_0 + mx)\,m$

(straight lines)
(with +ve slope & +ve intercept).

63)



A → B

$v' = a_1t_1$ ....(1)

$B \rightarrow C$

$v' = a_2t_2$ ....(2)

Solve (1) & (2)

$$\frac{t_1}{t_2} = \frac{a_2}{a_1} = \frac{6}{4} = \frac{3}{2}$$

64)

Newton's law of cooling :

$$\frac{\Delta T}{\text{time}} = K\left(\frac{\theta_1 + \theta_2}{2} - \theta_0\right)$$

$$\frac{61-59}{4} = K\left[\frac{61+59}{2} - 30\right] \quad ...(1)$$

$$\frac{51-49}{t} = K\left[\frac{51+49}{2} - 30\right] \quad ...(2)$$

On dividing, (1) & (2)

$$\Rightarrow \frac{t}{4} = \frac{30}{20} \Rightarrow t = 6 \text{ min.}$$

65) Breaking stress = $\frac{F}{A}$

$$\frac{1.5 \times 10^5}{\pi\left(\frac{3}{2}\right)^2} = \frac{F}{\pi\left(\frac{1.5}{2}\right)^2}$$

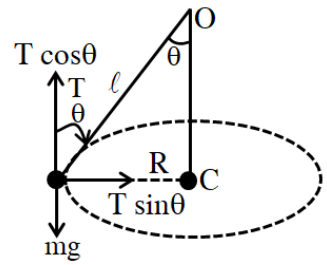
$$F = \frac{1.5 \times 10^5}{4}$$

$= 0.375 \times 10^5$ N

66)

$$T\sin\theta = \frac{mv^2}{R} \quad .....(1)$$

$T\cos\theta = mg$ ....(2)



Divide equation (1) by equation (2) : $\tan\theta = \frac{v^2}{Rg}$

$\Rightarrow v = \sqrt{Rg\tan\theta}$

$\Rightarrow \tan\theta = \frac{1}{\cot\theta}$

$\Rightarrow v = \sqrt{\frac{Rg}{\cot\theta}}$

67) $B^0 = \frac{E_0}{C}$

$= \frac{100}{3 \times 10^8}$

$= 0.33 \times 10^{-6}$ T

68) $E_{avg} = -L\frac{\Delta i}{\Delta t}$

$E_{avg} = -(5 \times 10^{-3})\left(\frac{10-20}{0.1}\right)$

$E_{avg} = 0.5$ V

69) If current leads voltage and angle between their phasors $< \frac{\pi}{2}$, it indicates that elements in circuit must contain R and C. It may contain L also, but then $X_C > X_L$. From given options, L is not there, so elements are R & C

70) $\frac{1}{\lambda} = \frac{1}{912\,\text{Å}}\left(\frac{1}{1^2} - \frac{1}{(3)^2}\right)$

$\frac{1}{\lambda} = \left(\frac{8}{9}\right)\left(\frac{1}{912\,\text{Å}}\right)$

$\lambda = \frac{9 \times 912}{8}$ Å = 1026Å

71) Neutral point will be near to the small current carrying wire and in outside at which

$\frac{\mu_0 9}{2\pi x} = \frac{\mu_0 \times 16}{2\pi(7+x)}$

$\Rightarrow$ x = 9 cm

 From wire of 16A current d = 7 + 9 = 16 cm

72) $B = \mu_0 nI$

$\Rightarrow \frac{B_1}{B_2} = \frac{n_i i_i}{n_2 i_2}$

$\Rightarrow \frac{6.28 \times 10^{-2}}{B_2} = \frac{200 \times \hat{i}}{100 \times i} \times 2$

$B_2 = \frac{6.28 \times 10^{-2}}{4} = 1.57 \times 10^{-2}$ Wb/m^2

73) M = N i A

$= 2 \times 0.5 \times \pi r^2$

$6.28 = (2\pi\, r) \times 2$

$r = \frac{6.28}{4\pi} = \frac{1}{2}$

$M = 1 \times \pi \times \frac{1}{4}$

$= \frac{\pi}{4}\ Am^2$

74) $\frac{\frac{1}{\rho}}{T}$ = constant

$\frac{1}{\rho T}$ = constant

ρT = constant

ρT ∝ P = constant

isobaric

$\frac{P}{\rho} = \frac{RT}{M}$

$\rho T = \frac{\rho T}{R}$

75) $\Delta W = nR\Delta T = 3 \times \frac{25}{3} \times 100 = 2500$ J

$\Delta U = nC_v\Delta T = 3 \times \frac{5R}{2} \times 100 = 6250$ J

Q = ΔU + W = 8750 J

$\frac{\Delta U}{\Delta W} = \frac{6250}{2500} = 2.5$

76) They are in same phase after n oscillation by large length pendulum and (n + 1) oscillation by small length pendulum.

 $(n + 1)\sqrt{\ell_s} = n\sqrt{\ell_e}$

⇒ $(n + 1)\sqrt{1} = n\sqrt{25}$

⇒ n + 1 = 5 n

⇒ 1 = 4n

$n = \frac{1}{4}$

 for smaller pendulum :

⇒ $n + 1 = 1 + \frac{1}{4} = \frac{5}{4}$

77)

$v = \sqrt{\frac{T}{\mu}}$

$v = \frac{w}{K} 30 m/s$

$30 = \sqrt{\frac{T}{0.16}}$

$30 = \sqrt{\frac{T \times 100}{16}}$

⇒ T = 144 N

78) Let R_1 = R, then R_2 = 2R

$$\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\frac{1}{6} = (1.5 - 1)\left(\frac{1}{R} - \frac{1}{-2R}\right)$$

R = R_1 = 4.5 cm

R_2 = 9 cm

79) $${}^2\mu_1 \times {}^3\mu_2 \times {}^4\mu_3 = \frac{\mu_1}{\mu_2} \times \frac{\mu_2}{\mu_3} \times \frac{\mu_3}{\mu_4} = \frac{\mu_1}{\mu_4}$$

$= {}^4\mu_1$

80) Value of g = $$g_s\left(1 + \frac{h}{R}\right)^{-2}$$

= $$g_s(1+2)^{-2} = \frac{g_s}{9}$$

Here g_s = gravitational acceleration at surface

Force = mg = 90 × $\frac{g_s}{9}$ = 100 N

82) Since only 6 different wavelength are excited, therefore highest excited state is n = 4.
Two wavelengths are shorter than λ_0, initially atoms were in excited state n = 2
Corresponding transitions are 4 → 3, 4 → 2, 4 → 1, 3 → 2, 3 → 1, 2 → 1.

83) B.E of Helium = $(2m_P + 2m_N - m_{He})\, c^2$
= 28.4 MeV

84) $$\lambda = \frac{h}{mv}$$

$\lambda_P = \lambda_a$

$m_P v_P = m_\alpha v_\alpha$

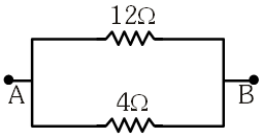
$m_P v_P = 4m_p v_\alpha$ $(m_\alpha = 4m_P)$

$$\frac{v_P}{v_\alpha} = 4$$

(Option 2) is correct

85) Measured value = 10 mm + 4(0.1 mm)
= 10.4 mm
= 1.04 cm

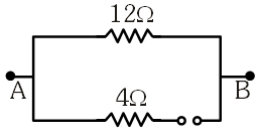
86) (i) $V_A > V_B$ ⇒ diode is in FB



⇒

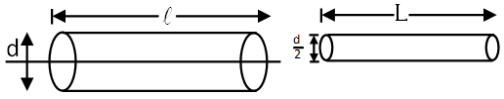
⇒ RAB = 3Ω

(ii) $V_A < V_B$ ⇒ diode is in RB



⇒

⇒ R_{AB} = 12Ω

87) 

Initial volume = final volume

$$\frac{\pi d^2 \ell}{4} = \frac{\pi d^2 L}{16} \quad \boxed{L = 4\ell} \quad A_f = \frac{A_i}{4}$$

$$R_i = \frac{\rho \ell}{A_i} \quad R_f = \rho\left(\frac{4\ell}{A_i}\right) = 16\, R_i$$

88) $f = \frac{1}{0.2}\text{cm}$

$$P = \frac{100}{f} = 100 \times 0.2 = 20\,D$$

89) $x_{cm} = \frac{A_1 x_1 - A_2 x_2}{A_1 - A_2}$

$A_1 = \pi(3R)^2 = 9\pi R^2$

$A_2 = \pi R^2$

$x_1 = 0$, $x_2 = 2R$

$$x_{cm} = \frac{0 - \pi R^2 \times 2R}{9\pi R^2 - \pi R^2}$$

$$x_{cm} = -\frac{R}{4}$$

90) The output (Y) = $\overline{\bar{A} \cdot \bar{B}}$ = A + B

BIOLOGY

91) NCERT-XI, Pg. # 32

92) NCERT Pg. No. # 27 & Module

93)

XI NCERT Pg. No. # 11

94)

NCERT Pg. No. 68

95) NCERT, Pg # 67
Both (1) & (2)

96) NCERT-XI, Pg # 146

97) NCERT, Pg. # 141

98)

NCERT-XI, Pg # 144

99) NCERT XI, Pg. # 150

100) NCERT-XI, Page No. # 144,146

101) NCERT XI Pg # 146

102) NCERT-XI, Pg # 139

103) NCERT-XI, Pg. 65

105)

NCERT-XI, Pg. No. # 14

106) NCERT-XII Pg. 12,13,14

107) NCERT XII Page No. 7

108) NCERT (XII) Pg # 9,10

109) NCERT XII Pg. # 6

110) NCERT XII Pg. # 12,18

111) NCERT-XII, Pg. # 67

112) NCERT XII Pg. # 60,61,69

113)

XII NCERT Pg # 59

114) NCERT-XII Pg. # 67, 70

115) NCERT-XII, Pg. # 59

116) NCERT-XII, Pg. # 63

117) NCERT XII Page No. # 70

118) NCERT-XII, Pg. # 61, 62

119) NCERT-XII, Pg # 95

120) NCERT-XII Pg # 86

121) NCERT-XII, Page No. # 88,89

122) NCERT-XII, Page No. # 91

123) NCERT-XII, Page No. # 83,85

124) NCERT-XII, Page No. # 94,95

125) NCERT-XII, Page No. # 91,93,95

126) NCERT-XII, Pg. # 103,105

127) NCERT XII Pg No. # 212

128)

NCERT-XII, Pg. No. # 207

129) NCERT XII Pg # 207

130) NCERT XII Pg. # 197

131)

Module

132) NCERT-XII Pg. # 201, 202

133) NCERT XII Pg # 196

134)

NCERT XII, Page No. 193

135)

NCERT XII, Pg. 222, 225

137) NCERT Pg, 160

139) NCERT Page No. 123

140) NCERT XII Pg # 117

141) NCERT Pg. # 48

142) NCERT, Pg # 44

143) NCERT Pg. # 46

144)

Module-6 Page No.#19

145) NCERT Pg. # 334

147) NCERT Pg. No. # 101

148) NCERT Pg. # 48

149) NCERT Pg. # 193

150) NCERT Pg. # 199

151) NCERT Pg # 35-36

152) NCERT, Pg. # 27

153) NCERT Pg. # (E)-37, (H)-41

154) NCERT, Pg. # 35

155) NCERT-XI Pg. # 317

156) NCERT Pg#236

157) NCERT-XI Pg#304, Ist para

158) NCERT, Pg. # 310

163) NCERT XII Pg # 199

164)

NCERT-XII, Pg. # 196

165)

NCERT-XII, Pg. # 184,185

167) NCERT (XIth) Pg. # 168 : 10.4.1

168)

NCERT-XI, Pg # 137 and 138

169) NCERT-XI, Pg # 143

173) NCERT Pg. # 114

175) NCERT Pg. # 243

176) NCERT Pg. # 244

177) NCERT, Pg. # 43

180) NCERT Pg. # 208